Recoverability: definitions, formulations and a constructed CARLA case catalogue

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Purpose. Replace an open-ended search for interesting recorded episodes with a small, explicitly defined library of mechanisms. Each case states what should make recovery possible or impossible, how to instantiate that mechanism in CARLA, and what evidence is needed before assigning a label.

Status. This report specifies proposed experiments. No CARLA scenario has been executed or validated for this report. Numerical values are starting points or analytical examples, not measured vehicle capabilities. Case intentions must never be copied directly into training labels.

The central definition

A history is recoverable when at least one executable, observation-conditioned continuation policy satisfies the declared constraints across the considered worlds and reaches a supported terminal continuation. Recoverability depends on the vehicle, available information, delays, environment assumptions, policy class and evaluation horizon; it is not a property of an image alone.

Three distinctions that govern the benchmark

Do not equate	Use instead
A tested maneuver fails	That particular policy is refuted; other policies may work.
A search finds no solution	No witness was found in that search, not physical impossibility.
The first obstacle is avoided	The full footprint remains valid through stabilization and post-maneuver continuation.

Source basis and proposed extensions

The attached proposal [1, Sections 4–6] supplies the history-based formulation, finite causal-policy library, recovery profiles, terminal checking and unresolved-evidence category. This report preserves those semantics. Its evidence hierarchy, twenty case definitions, parameter rules, construction workflow and data manifest are proposed benchmark specifications, not results claimed by the proposal.

Inevitable-collision-state research supplies the stronger notion of unavoidable collision under declared dynamics [2]. Research on occlusion-aware control motivates distinguishing physical possibility from what a partially informed agent can implement [3]. Official CARLA documentation supports the implementation notes [4–12]. These sources do not certify the cases defined here.

Deliverables and reading map

Sections 1–4 define the mathematical and evidence contracts. Section 5 indexes the twenty cases; Section 6 gives their construction cards. Sections 7–11 specify CARLA execution, labeling, evaluation and build priorities. A machine-readable catalogue and an illustrative project-specific YAML manifest accompany the report. They are specifications, not runnable OpenSCENARIO files.

1. Define the contract before the label

A change of assumptions can legitimately change recoverability.

For every benchmark instance, record the contract below. An unqualified “recoverable/unrecoverable” label is incomplete without this information. The source proposal explicitly makes its values relative to a policy library, reference worlds, dynamics, horizon and margin threshold [1, pp. 2–3].

Element	Required declaration
Dynamics F	Ego and actor dynamics; control bounds; tracking errors; actuator state; vehicle footprint; physics configuration.
Observation model O	Sensors or structured observations, visibility masks, timestamp semantics, delay, noise and retained history.
World set Ω(h)	Hidden states, actor response mechanisms and disturbances compatible with the observed history. Finite support first.
Policy class Π	Executable feedback policies; parameters; observable branch predicates; action durations and unavailable actions.
Constraints and threshold	Collision, permitted road area, physical limits and terminal requirements; units; normalization; threshold ε.
Timing and terminal rule	Evaluation horizon, continuation duration, decision/actuation timing and the exact recovery-completion predicate.

State is richer than pose

Let x include ego position, heading, speed, yaw rate and relevant actuator state; actor states; road geometry; simulation time; and queued observations/commands when delays matter. A simple reference evolution is x(k+1) = F(x(k), u_applied(k), ω). Including queues prevents two states with identical poses but different pending commands from being incorrectly treated as equivalent.

History is what the policy was actually allowed to know

$$h_t = \left\{ \left(o_i, t_i^{capture}, t_i^{available} \right) : t_i^{available} \le t \right\} \cup \left\{ \text{past commands and applied controls} \right\}$$

Interpret h as a time-ordered record, not an unordered feature bag. A controller may use only available data and its retained memory. A frame captured earlier but delivered later is not available retrospectively. The evaluator may read simulator ground truth, but the policy input must not contain hidden mode IDs, future trajectories, evaluator outcomes or scenario names.

A world is a mechanism, not a prerecorded path

A world fixes relevant initial conditions and actor-response rules. An actor may brake or yield differently after different ego actions. Across candidate evaluations, keep the same mechanism and exogenous variables, not necessarily the same realized actor trajectory. Each world must remain compatible with the common pre-branch history. Changing future ego actions must not rewrite past observations.

2. Mathematical formulation

Robust recovery is an existence claim about an implementable policy.

2.1 Per-policy constraint margin

Let m_i(x_s) be signed margins for hard trajectory constraints, positive on the accepted side. Let q_i > 0 normalize their units. Let m_term evaluate the specified terminal continuation. The complete rollout includes the emergency action, stabilization and the post-maneuver interval.

$$G(\pi,\omega;h)=\min\left\{\min_{s\in[t,t+H)}\min_{i}\frac{m_i\left(x_s^{\pi,\omega}\right)}{q_i},\frac{m_{term}(\pi,\omega;h)}{q_{term}}\right\}.$$

$$r_\pi(h)=\min_{\omega\in\Omega(h)}G(\pi,\omega;h),\ R_\Pi(h)=\left[r_{\pi_1}(h),\ldots,r_{\pi_N}(h)\right).$$

If the terminal test is Boolean, define G as the path margin on a pass and as negative infinity on a failure. Store the terminal Boolean separately, not as a fabricated distance. G and R are reference quantities, not collision probabilities. This expands the normalized-margin construction in [1, pp. 2–3].

2.2 Recoverability and unrecoverability at a declared scope

$$Rec(h;\Pi,\Omega)\Leftrightarrow\exists\pi\in\Pi_{causal}\,\forall\omega\in\Omega(h):G(\pi,\omega;h)\geq\varepsilon.$$

$$Unrec(h;\Pi,\Omega)\Leftrightarrow\forall\pi\in\Pi_{causal}\,\exists\omega\in\Omega(h):G(\pi,\omega;h)<\varepsilon.$$

Start with policies deterministic given history. Randomized policies require explicit treatment of their randomness. A positive label needs a common causal policy across the worlds. A negative label needs the universal claim over the declared policy class. Exhaustive finite-library failure establishes only that library's failure. A claim over all dynamically admissible policies requires an additional proof or sound verifier.

2.3 Action-conditioned recovery

$$V_\Pi(h,a)=\max_{\pi\in\Pi:\pi\,begins\,with\,a}r_\pi(h),\ A_{rec}(h)=\{a:V_\Pi(h,a)\geq\varepsilon\}.$$

Specify the duration of the initial action; "steer left" is not a reproducible prefix. For finite Π the maximum is well-defined. For a continuous class, a supremum at the threshold does not by itself establish an attained feasible policy. Treat unresolved boundary cases accordingly.

2.4 Physical possibility is not information-feasible recovery

$$\text{Per-world feasibility: }\forall\omega\,\exists\pi_\omega:G(\pi_\omega,\omega;h)\geq\varepsilon.$$

$$\text{Causal recovery: }\exists\pi_{causal}\,\forall\omega:G(\pi_{causal},\omega;h)\geq\varepsilon.$$

The first statement does not imply the second. Each hidden world can have a successful hindsight-specific action while no common observation-constrained policy succeeds. This is robust information infeasibility, not proof that collision occurs in every actual realization. Chance-constrained success under an explicit probability distribution is a separate objective; do not silently substitute it for the robust definition.

3. Evidence, constraints and terminal behavior

The label must say both what is claimed and why it is supported.

Recorded outcome	Evidence required and permitted interpretation
R_REF	A witness policy satisfies every declared finite world and the terminal test. Record whether evidence is analytical, model-exhaustive or executed finite replay.
U_LIBRARY	Every member of the named finite policy library has a valid refutation. No conclusion about policies outside that library.
U_MODEL	A sound argument excludes every admissible policy in the specified reference dynamics and horizon. Not automatically a certificate for CARLA or the real vehicle.
U_INFORMATION	A qualified U_MODEL or U_LIBRARY result caused by delayed/missing information, with per-world solution witnesses. Store the scope explicitly.
UNRESOLVED	Search is incomplete, a policy/world entry is missing, errors overlap the threshold, or transfer validity is unknown. Do not use as a negative label.
INVALID_RUN	Spawn, site geometry, controller, timing, world compatibility or logging contract failed. Exclude from safety-performance denominators.

A counterexample refutes a particular policy, not the entire scene. A successful rollout in one world supports only that execution unless the remaining declared worlds are covered. A learned prediction receives none of the above evidential authority by itself. These distinctions implement the proposal's witness/counterexample/unresolved rule [1, p. 3].

Hard constraints must be separated from preferences

Use full-body collision avoidance, explicitly permitted road area and physical actuation limits as hard constraints. For geometry, a usable margin is signed body clearance minus a declared positive reserve; a zero margin then means meeting that reserve, not accepting contact. Store route progress, preferred lane, comfort and task completion separately unless the task explicitly makes one a hard requirement. Failure to complete the original route is not automatically loss of collision-avoidance recoverability. Name the failed specification.

A supported terminal continuation is not merely zero speed

Define a terminal mode such as protected hold with a specified future actor response, or stable lane following with checked spacing and tracking. Verify the full post-maneuver interval. A stop in the path of an approaching rear actor is not accepted just because ego speed is zero. A finite checked continuation supports only its stated horizon; indefinite safety needs a separately justified invariant set. The attached proposal requires this distinction [1, p. 2].

Numerical uncertainty and already-violated starts

Where a verifier supplies justified bounds [lower, upper] on r, accept only if lower ≥ ε and refute only if upper < ε. Otherwise record UNRESOLVED. Repeated-run ranges are empirical sensitivity evidence, not certified bounds. Exclude spawn overlaps. If a hard constraint is already violated at t_0, the zero-violation recovery definition cannot undo it; report any later damage-mitigation task separately.

4. Boundary calculations and information timing

Use derivable boundaries to construct cases, then measure the CARLA realization.

4.1 An exact one-dimensional reference example

Assume a level one-dimensional corridor, initial speed v, fixed delay τ during which speed remains constant, maximum achievable deceleration b > 0 afterwards, no lateral escape, no rear threat, and a stationary obstacle. Distances are bumper-to-obstacle clearances. These are analytical assumptions, not default CARLA properties.

$$D_{stop} = v\,\tau + \frac{v^2}{2b}, d_{required} = D_{stop} + c\,.$$

With v = 12 m/s, τ = 0.25 s, b = 6 m/s² and desired reserve c = 1 m: D_stop = 15 m and d_required = 16 m. A gap of 18 m admits the ideal maximum-braking witness with 3 m final clearance. A gap of 12 m leads to contact before stopping under this model. A gap of 15.5 m avoids contact ideally but fails the 1 m reserve. Keep these three labels distinct.

The impossibility argument uses a genuine upper bound on total achievable deceleration; the positive witness uses achievable performance. Do not use a comfortable braking value to prove impossibility, or an occasional best stopping result as guaranteed braking. Nonzero acceleration during the delay requires integrating that pre-response motion. A full-width physical blocker is needed for an inevitable-collision claim; a painted road boundary alone establishes only constraint infeasibility [2].

4.2 When can observation-dependent recovery work?

$$t_{available} + \tau_{planning} + \tau_{command/actuation} \leq t_{latestbranchresponse}.$$

The right-hand side is the latest effective branch-response time for the tested continuation. Compute it from execution or the reference dynamics. Avoid double-counting overlapping measured delays. Early evidence does not guarantee success; all branches, the common prefix and terminal behavior still require checks. Delayed evidence yields an information-infeasible case only when no shared stopping, waiting, straight-through or other supported response exists [3].

4.3 Loss of recoverability is policy-dependent

Fix a baseline prefix β, such as continuing at the current speed. Evaluate recovery after different durations of that prefix while checking the prefix itself. This defines a set of admissible intervention delays. Moving actors can create several feasible intervals, so do not assume every delay sweep is monotonic. Use bisection only after validating monotonicity for the particular family; otherwise perform a coarse scan followed by local refinement.

Treat distance, response delay, gap phase and reveal time as the main construction axes. The goal is a small number of audited positive, boundary and negative anchors—not a blind search through unrelated recorded scenes.

5. Catalogue index

Twenty definitions: eighteen mechanism/stress cases and two explicit controls.

All entries are **proposed construction targets**. R_REF, U_LIBRARY and U_MODEL are awarded only after the stated evidence checks. A false rejection or false admission is a relationship between a model's judgment and supported reference evidence; it is not a label assigned from visual appearance.

ID	Defined case	Primary contrast
RC01	Braking with a protected stop	Braking witness / gap boundary
RC02	Too late to stop in a closed corridor	Model impossibility / open escape
RC03	Braking fails but lateral escape remains	Brake-only / lateral recovery
RC04	Staged slowing before a feasible lane change	Single action / staged response
RC05	Aborting an unsafe lane change	Commit / abort
RC06	Yield, then use a time-varying gap	Immediate / later gap
RC07	Primary avoidance with continued secondary-threat management	First avoidance / continuation
RC08	Protected waiting despite route noncompletion	Safety / route progress
RC09	Front/rear conflict with no supported escape	Endpoint failures / exhaustive class
RC10	Early evidence enables a contingent escape	Fixed plan / causal branching
RC11	Late evidence prevents robust causal recovery	Hindsight / available information
RC12	Recoverability lost by delaying intervention	Early / delayed intervention
RC13	Over-approximated predictions falsely close a corridor	Credible worlds / approximation
RC14	Current-view aliasing resolved by past observations	Current frame / history
RC15	Yielding must be observed, not assumed	Assumed yield / reactive worlds
RC16	Center-point clearance but body-footprint conflict	Point / swept footprint
RC17	Response delay eliminates an earlier recovery option	Ideal / realized response time
RC18	Requested control exceeds available authority	Requested / executable control
RC19	Benign and label-preserving nuisance controls	Semantic / nuisance variation
RC20	Incomplete library must remain unresolved	No witness / no possible policy

The cases intentionally share infrastructure. RC01/02/12/17 share a stopping corridor; RC03/04/05/06/07/15/16/18 share a two-lane site; RC10/11/14 share observation-control machinery. RC13 changes a predictor approximation, and RC20 changes evidence coverage, not the underlying physical scene. Reuse these contrasts rather than claim twenty unrelated road geometries.

6. Case cards | RC01–RC02

Project-defined cases. Starting parameters are not CARLA-validated labels.

RC01 Braking with a protected stop

Site / intended evidence: Straight constrained corridor. R_REF.

Mechanism. A stopped obstacle is visible early enough for an executable braking policy to stop and remain protected.

CARLA construction. Place ego and a full-width stationary blocker on a straight site. Keep the rear and terminal hold area free of relevant threats. Define t_0 from the measured bumper clearance after a warm-up approach, not from spawn time.

Parameters. Pilot: v = 12 m/s; gap near 18 m only for the analytical contract on page 5. In CARLA choose gap = measured conservative stopping distance + 1–3 m reserve. Sweep one variable: gap.

Witness or refutation. Witness: brake within the calibrated envelope, stabilize, then hold for the declared continuation interval.

Accept only when. No swept-footprint conflict, road or dynamics violation; positive clearance reserve throughout; final hold passes. Record an analytical witness separately from CARLA execution support.

Paired contrast. RC02 decreases gap while preserving vehicle, delay and road contract. Boundary variants separate contact avoidance from reserve compliance.

Required profile contrast. r(brake_hold) >= ε; terminal hold passes.

Research diagnostic. Basic positive control and a calibration target. A braking test alone is not a complete recoverability benchmark.

RC02 Too late to stop in a closed corridor

Site / intended evidence: Same corridor as RC01. U_MODEL in the analytical model; CARLA scope must be qualified.

Mechanism. An obstacle blocks every admissible route before the earliest possible stop.

CARLA construction. Reuse RC01. Use a collidable full-width obstruction and validated side boundaries; exclude a reachable shoulder, junction or gap. Initialize before any collision. Do not disable physically allowed steering merely to claim real-world impossibility.

Parameters. Analytical anchor: v = 12 m/s, τ = 0.25 s, b_max = 6 m/s², gap = 12 m. Sweep gap around the stopping boundary. Replace b_max with a justified bound for the named reference dynamics.

Witness or refutation. Refutation: stopping-distance lower bound plus proof that no alternative route crosses the blocked section. Finite controller failures alone yield U_LIBRARY at most.

Accept only when. The infeasibility proof must cover all admissible controls of the declared model. In CARLA, confirm layout and calibrated response; keep the model certificate and empirical execution evidence separate.

Paired contrast. Increase gap to RC01, or open a physically reachable escape lane to RC03.

Required profile contrast. Every admissible model policy violates the stated threshold; collision inevitability is a separate stronger predicate.

Research diagnostic. A clean negative anchor. Impact-speed reduction is mitigation, not a recovery witness.

6. Case cards | RC03–RC04

Project-defined cases. Starting parameters are not CARLA-validated labels.

RC03 Braking fails but lateral escape remains

Site / intended evidence: Two-lane straight road. R_REF if an executable bypass witness exists.

Mechanism. The longitudinal stopping requirement is not met, but a supported adjacent corridor permits evasive motion.

CARLA construction. Place a stationary blocker only in ego's lane. Keep one adjacent lane free through the complete maneuver and stabilization interval; block the opposite side only if part of the declared geometry.

Parameters. Pilot speeds 8–14 m/s. Derive obstacle distance from the measured lateral maneuver, not from an assumed lane-change time. Find a gap where braking fails but the executed bypass passes.

Witness or refutation. Witness: a calibrated steer or brake–steer continuation followed by stable tracking. Use a complete policy, not an untracked geometric spline.

Accept only when. Replay braking to refute it and the bypass to support it across all declared worlds. Check footprint, steering rate, lateral acceleration and downstream traffic.

Paired contrast. Close the escape corridor without changing the front obstacle; do not prelabel the resulting scene U_MODEL.

Required profile contrast. r(brake_hold) < ε and r(executable_bypass) >= ε.

Research diagnostic. Tests false rejection from one-dimensional or braking-only reasoning. This is a policy-set distinction, not automatically a latent-model failure.

RC04 Staged slowing before a feasible lane change

Site / intended evidence: Two-lane straight road. R_REF; single-stage candidates may fail.

Mechanism. An immediate high-speed lane change violates clearance or dynamic limits; braking in-lane first reduces speed enough for a later executable change.

CARLA construction. Use a front obstacle and a narrow but usable neighboring corridor. Give enough initial distance for the staged action. Start from a calibrated low-speed maneuver and construct the preceding braking segment backward.

Parameters. Sweep brake-prefix duration 0.2–1.0 s initially, then lateral-maneuver duration and front gap. Treat all values as search starts. Coupled braking/steering must respect the measured combined envelope.

Witness or refutation. Witness: moderate braking, transition under declared guards, then lane change and stabilization. Straight braking and the chosen immediate evasive candidates provide policy-specific counterexamples.

Accept only when. The full staged controller must execute without saturation-induced tracking failure. A path assembled from individually feasible segments is insufficient.

Paired contrast. Remove the necessary prefix from the library for RC20; increase delay or shorten distance for RC17.

Required profile contrast. r(staged_brake_change) >= ε; named immediate candidates < ε.

Research diagnostic. Tests sequencing and latent rollout consistency; useful for held-out policy compositions.

6. Case cards | RC05–RC06

Project-defined cases. Starting parameters are not CARLA-validated labels.

RC05 Aborting an unsafe lane change

Site / intended evidence: Two parallel lanes. R_REF through abort or reselection.

Mechanism. The intended target gap closes while a return to the original corridor remains feasible.

CARLA construction. Initialize ego during a physically executed lane-change prefix, not by teleporting to an arbitrary lateral pose. Trigger the target-lane actor's response at an observable event. Preserve adequate source-lane space.

Parameters. Sweep lateral progress at hazard reveal, target-actor relative speed, source-lane rear gap and control latency. Start well before the abort boundary.

Witness or refutation. Witness: cancel the merge, return under steering-rate limits, adjust speed and stabilize. The original committed lane change is separately refuted.

Accept only when. Evaluate both target and source lanes during the return. The abort cannot depend on an unobserved actor-mode ID.

Paired contrast. Add a source-lane blocker or delay observation. Failure of abort does not exclude completing the merge or another policy.

Required profile contrast. r(commit_original_merge) < ε; r(abort_and_stabilize) >= ε.

Research diagnostic. Distinguishes task/maneuver failure from loss of all recovery options.

RC06 Yield, then use a time-varying gap

Site / intended evidence: Two-lane straight road or merge. R_REF via a timed or observation-conditioned continuation.

Mechanism. Immediate entry conflicts with a passing vehicle, but waiting in a supported state creates an admissible later gap.

CARLA construction. Place a passing actor in the adjacent lane. Provide enough longitudinal reserve to slow or hold while it clears. Its acceleration law and downstream motion must be explicit.

Parameters. Sweep relative longitudinal phase, passing speed, waiting-room distance and gap-observation delay. Do not assume time-to-feasibility is monotonic.

Witness or refutation. Witness: remain supported, observe actual clearance, then enter behind the passing vehicle. If a known timed plan suffices, record it rather than claiming feedback is necessary.

Accept only when. Check safety during waiting and after the merge. Positive evidence cannot rely on the actor yielding unless that response belongs to the declared world set.

Paired contrast. Shorten waiting room or add a non-clearing actor mode. The robust label can change even when one favorable world still succeeds.

Required profile contrast. r(immediate_entry) < ε; r(wait_observe_enter) >= ε.

Research diagnostic. Tests time-space reasoning and false rejection by instantaneous-gap tests.

6. Case cards | RC07–RC08

Project-defined cases. Starting parameters are not CARLA-validated labels.

RC07 Primary avoidance with continued secondary-threat management

Site / intended evidence: Two-lane road with rear actor. R_REF only with complete post-maneuver policy.

Mechanism. Two policies avoid the front obstacle, but one hands back or stops in the path of a fast rear actor. A continued controlled response remains feasible.

CARLA construction. Use a front blocker and a rear/adjacent actor arriving after the first avoidance. Keep a verified continuation corridor available. Position the secondary conflict after the short nominal scoring horizon.

Parameters. Sweep secondary-actor arrival time, handback time and stabilization duration. Initialize far enough inside the positive region to establish a robust full-horizon witness.

Witness or refutation. Witness: complete primary avoidance and continue in the supported corridor until rear separation and tracking satisfy the terminal contract.

Accept only when. Compare short-horizon success with full-horizon margins. Refute the premature-handback policy without falsely declaring the scene unrecoverable.

Paired contrast. Close the continuation corridor or advance the rear actor; the negative side requires complete library evaluation or a stronger proof.

Required profile contrast. Primary-only test passes; full handback policy < ε; supported continuation >= ε.

Research diagnostic. Tests terminal supervision and secondary hazards, directly matching the selected proposal.

RC08 Protected waiting despite route noncompletion

Site / intended evidence: Straight road with temporary blockage. R_REF for safety; route success may be false.

Mechanism. The original goal cannot be reached during the task window, but a protected hold remains executable.

CARLA construction. Place a temporary blockage beyond a valid stopping location. Specify rear-actor behavior and whether the blockage clears. Make safety and route-completion predicates separate fields.

Parameters. Sweep blockage duration across the route deadline; retain the same checked hold interval. A hold interval of 3–5 s is a starting test duration, not indefinite safety.

Witness or refutation. Witness: decelerate into the protected set and maintain it. For later route completion, evaluate a separate restart policy after observed clearance.

Accept only when. Accept only when the hold is physically supported against the reference worlds. Label collision-avoidance recovery separately from task success.

Paired contrast. Add a rear actor that invalidates the hold. Zero speed must not override the failed continuation margin.

Required profile contrast. r(protected_hold) >= ε while the separately stored route goal is not completed.

Research diagnostic. Tests false rejection caused by treating progress or nominal route completion as a safety constraint.

6. Case cards | RC09–RC10

Project-defined cases. Starting parameters are not CARLA-validated labels.

RC09 Front/rear conflict with no supported escape

Site / intended evidence: Physically bounded corridor. U_LIBRARY unless a full-model proof is available.

Mechanism. A front obstacle demands slowing while a rear actor's approach makes certain braking responses unsafe; lateral escape is unavailable.

CARLA construction. Use full road geometry, front blocker and explicit rear response dynamics. Confirm that the initial footprints do not overlap and that actor reactions are causal.

Parameters. Sweep front clearance, rear clearance, rear closing speed, rear braking delay and ego response bounds. Begin from known per-policy refutations, not an assumed global negative.

Witness or refutation. For U_LIBRARY, provide a counterexample for every finite causal policy, including intermediate braking and allowed forward progress. For U_MODEL, use a valid reachable-set or interval argument covering all controls.

Accept only when. Reject a negative construction if any intermediate action or timing creates a supported window. Failure of "hard brake" and "full throttle" is not exhaustive.

Paired contrast. Open an escape corridor or improve rear response, then obtain a positive witness.

Required profile contrast. For every policy in the stated class, at least one credible world supplies a refutation.

Research diagnostic. Hard negative with strong evidence requirements; prioritize only after simple negatives work.

RC10 Early evidence enables a contingent escape

Site / intended evidence: Obstacle with two downstream escape corridors. R_REF under the information-constrained contract.

Mechanism. A hidden actor mode blocks one of two escape corridors. A shared initial slowdown preserves enough time to observe the mode and choose the other corridor.

CARLA construction. Create two worlds with identical permitted histories before reveal. Use an observation gate initially; a physical occluder can follow. Ensure stopping in place and every fixed full maneuver fail in at least one world if claiming feedback necessity.

Parameters. Sweep reveal time relative to the measured last branch-response time. Preserve the same physical worlds across early/late observation variants. Require a positive timing reserve.

Witness or refutation. Witness: a common safe prefix, a branch triggered by delivered observations, and a supported continuation in each mode.

Accept only when. Unit-test identical pre-reveal action traces. Verify each branch, latency, full footprint and terminal behavior. A fixed timed solution invalidates the claim that contingency is necessary, though the scene can still be recoverable.

Paired contrast. RC11 delays only information delivery beyond the effective branch deadline.

Required profile contrast. r(causal_branch) >= ε; fixed-plan necessity requires refuting the named open-loop class.

Research diagnostic. Tests causal policies rather than hindsight trajectories. An ordinary intersection where stopping works is not a valid necessity example.

6. Case cards | RC11–RC12

Project-defined cases. Starting parameters are not CARLA-validated labels.

RC11 Late evidence prevents robust causal recovery

Site / intended evidence: Same physical layout and worlds as RC10. U_INFORMATION at the explicitly proved scope.

Mechanism. Each hidden world remains individually solvable, but the required decisions differ before the policy can distinguish them.

CARLA construction. Keep RC10 dynamics and initial worlds fixed. Delay only the permitted observation stream. Use an exactly enumerated finite information tree before attempting a continuous-policy claim.

Parameters. Place evidence arrival after the last common response opportunity. Check all available shared actions: stopping, waiting, continuing and both escapes.

Witness or refutation. Provide per-world solution witnesses plus a proof that every shared causal policy in the named class is refuted in at least one world. A complete finite tree supports only that discrete reference class.

Accept only when. If a common stop or other robust prefix succeeds, this is not an information-infeasible case. Do not describe zero robust success as collision in every realized world.

Paired contrast. Reveal the same evidence earlier to recover RC10. Geometry and hidden-mode priors do not change.

Required profile contrast. Every world has a witness, but no shared causal policy passes within the proved class.

Research diagnostic. Detects clairvoyant teachers and hidden-state leakage. Distinct from physical unavoidable collision.

RC12 Recoverability lost by delaying intervention

Site / intended evidence: Reuse RC01, RC03 or RC07. Positive before a measured boundary; scope-qualified negative afterward.

Mechanism. A collision-free baseline action consumes the time or space needed by the recovery policy.

CARLA construction. Execute a fixed baseline prefix, such as maintaining speed, for different durations before handing over. Reconstruct the same initial world and retain identical actor mechanisms.

Parameters. Sweep prefix length in ticks, then refine around a validated crossing. Keep event deadlines and terminal contract comparable; do not shorten post-maneuver checking in later interventions.

Witness or refutation. Find a supported witness before loss. After loss, state whether the evidence is U_LIBRARY, U_MODEL or UNRESOLVED.

Accept only when. Record the feasible-delay set. Do not force a single threshold when moving traffic creates disjoint windows. Check the entire baseline prefix as well as the recovery.

Paired contrast. Use matched early/late snapshots from the same family and group them in one data split.

Required profile contrast. The supported action set changes along the declared baseline prefix; record all observed feasible intervals.

Research diagnostic. Tests preservation of options before immediate collision risk becomes high.

6. Case cards | RC13–RC14

Project-defined cases. Starting parameters are not CARLA-validated labels.

RC13 Over-approximated predictions falsely close a corridor

Site / intended evidence: Reuse a crossing or merge site. R_REF with a validated witness under the declared dynamics.

Mechanism. A deliberately inflated or kinematically inconsistent prediction envelope rejects an action whose true reference-world continuations are supported.

CARLA construction. Keep the physical scene fixed. Construct a pessimistic baseline predictor that violates an explicit actor speed bound or retains impossible cross-time mode switches. Evaluate against coherent reactive worlds.

Parameters. Vary envelope inflation and time resolution without changing ground-truth actor laws. Record exactly which included states are unreachable under the reference contract.

Witness or refutation. A causal witness must pass every credible reference world. Provide evidence that the added envelope region, rather than a genuine hazard, causes baseline rejection.

Accept only when. A union of valid per-time occupancies is not automatically incorrect for an open-loop collision test. Do not remove credible worlds merely to obtain a positive label.

Paired contrast. Replace the approximate envelope with a coherent representation or introduce a genuinely faster actor as a distinct physical variant.

Required profile contrast. r(witness) >= ε in coherent worlds; the approximate predictor rejects it.

Research diagnostic. Tests false rejection due to approximation. Separate this from RC10's need for feedback branching.

RC14 Current-view aliasing resolved by past observations

Site / intended evidence: Occluding van or junction. Different history-conditioned profiles; current frame may be identical.

Mechanism. Two current observations look the same, but earlier observations established different hidden-actor positions or motion.

CARLA construction. Generate two physically compatible approach histories. Reveal the actor earlier, then occlude it at t_0. Supply the complete permitted history to the memory model and only the final frame to the diagnostic baseline.

Parameters. Sweep time since last observation and motion uncertainty. Keep actor IDs or color from acting as unintended hidden-mode labels. History length is explicitly recorded.

Witness or refutation. Evaluate the recovery profile conditional on each history and its compatible worlds. Where the full histories are truly identical, use one belief-conditioned target rather than contradictory labels.

Accept only when. Confirm that the decisive evidence occurs before t_0 and remains causally available. Revalidate worlds after every observation mask change.

Paired contrast. Mask the earlier observation and recompute Ω(h); this changes information, not physical state.

Required profile contrast. Current-view features match, full histories differ, and their compatible recovery profiles differ.

Research diagnostic. Tests persistent memory and distinguishes missing perception from lost learned representation.

6. Case cards | RC15–RC16

Project-defined cases. Starting parameters are not CARLA-validated labels.

RC15 Yielding must be observed, not assumed

Site / intended evidence: Two-lane merge. R_REF only if a common causal fallback survives both modes.

Mechanism. A merge works when the adjacent actor yields but fails if that favorable response is assumed in a non-yielding world.

CARLA construction. Define yield and non-yield modes with bounded reaction delay and acceleration. Both respond to actual ego state. Provide a pre-commitment slowdown or wait that keeps a fallback executable.

Parameters. Sweep actor response delay, rear gap, ego commitment time and response observability. Maintain the same exogenous mode across candidate-policy comparisons.

Witness or refutation. Witness: observe a sufficient gap before merging; otherwise remain in the supported fallback. The always-merge policy supplies a non-yield counterexample.

Accept only when. The common fallback must pass before the mode is known. A positive label from a recorded yielding trajectory alone is invalid.

Paired contrast. Remove fallback room or reveal the response too late; preserve separate labels for favorable-world success and robust success.

Required profile contrast. r(always_merge) < ε; r(observe_yield_else_fallback) >= ε.

Research diagnostic. Tests interactive world models and false feasibility caused by replaying cooperative behavior.

RC16 Center-point clearance but body-footprint conflict

Site / intended evidence: Tight bypass or narrow gate. Specific maneuver refuted; scene may still be R_REF.

Mechanism. A geometric centerline clears an obstacle while the vehicle's swept body or overhang intersects it.

CARLA construction. Place a narrow gate or offset blocker on a bypass route. Use the actual ego bounding-box offset and rotation; inspect front and rear swing during the turn.

Parameters. Sweep gate width and lateral offset around footprint clearance; vary vehicle length only as a separate physics/geometry condition, not a cosmetic nuisance.

Witness or refutation. Refute the point-safe maneuver with a body-level trace. Search for a supported alternative such as earlier braking or a wider path before assigning a scene label.

Accept only when. Use consistent timestamps and collision-event corroboration. A conservative OBB overlap is a reference-geometry violation, not automatically a detailed mesh contact.

Paired contrast. Widen the gate or change the approach with a verified alternative.

Required profile contrast. Point-path clearance is positive while a body-level constraint for that maneuver fails.

Research diagnostic. Tests geometry fidelity; one invalid trajectory must not become a global negative.

6. Case cards | RC17–RC18

Project-defined cases. Starting parameters are not CARLA-validated labels.

RC17 Response delay eliminates an earlier recovery option

Site / intended evidence: Reuse braking or bypass site. R_REF to negative/unresolved according to the named scope.

Mechanism. A policy succeeds with a short delivered-observation-to-response delay and fails when more delay is introduced.

CARLA construction. Insert explicit observation and command queues. During the delay, continue the declared prior control; do not freeze the world. Use the same site, actors and controller parameters.

Parameters. Start with delays of 0, 0.15 and 0.30 s on a compatible tick grid, then refine. Log realized delays rather than relying on configuration alone.

Witness or refutation. Compare identical policy executions; search the remaining class before labeling the higher-delay scene unrecoverable. The corridor variant can use the analytical bound in Section 4.

Accept only when. No backdated sensor packets; no fictitious instantaneous braking. Keep sensor capture, delivery, decision, command application and measured response distinct.

Paired contrast. Shorten delay without altering physical worlds. Alternatively enlarge gap as a separate contrast.

Required profile contrast. The same policy passes at the shorter measured delay and fails at the longer delay.

Research diagnostic. Tests world-model timing and the gap between proposal feasibility and executed control.

RC18 Requested control exceeds available authority

Site / intended evidence: Curved bypass or combined brake–steer site. Specific command refuted; scene label requires further evidence.

Mechanism. A predicted path requires steering rate, yaw rate, deceleration or combined force beyond the declared envelope.

CARLA construction. Use a qualified low-demand maneuver and progressively increase demand or reduce measured authority. Keep scored motion physics-enabled. Record applied controls and actual response.

Parameters. Sweep turn duration, entry speed and combined braking demand. Change wheel/vehicle physics only through a versioned profile that is recalibrated.

Witness or refutation. Refute over-demanding commands; seek a staged or lower-speed supported continuation. A simplistic friction-circle estimate is only a screening model unless validated.

Accept only when. Do not assume max brake and max lateral acceleration are jointly attainable. Weather appearance alone must not be used as evidence that tire capability changed.

Paired contrast. RC04 provides a potential staged positive; reduce demand while preserving the scene.

Required profile contrast. Geometric candidate passes but its executed dynamics constraint fails; assess other policies separately.

Research diagnostic. Tests dynamic feasibility and prevents geometric paths from being mislabeled as executable recovery.

6. Case cards | RC19–RC20

Project-defined cases. Starting parameters are not CARLA-validated labels.

RC19 Benign and label-preserving nuisance controls

Site / intended evidence: Any qualified site. R_REF for benign anchor; paired profiles invariant only when justified.

Mechanism. The model should preserve ordinary progress and should not infer recovery labels from scenario appearance.

CARLA construction. Create a clear-road or non-conflicting-actor anchor. Separately rerender a qualified family with nuisance changes that leave dynamics, relevant observations and policy semantics unchanged.

Parameters. Candidate nuisances: paint/material changes or distant noninteracting objects after visibility checks. Do not treat darkness, fog, vehicle size, road friction or visibility changes as automatically invariant.

Witness or refutation. Nominal lane following supports the benign anchor. For nuisance pairs, verify physical profiles and information equivalence rather than assuming invariance.

Accept only when. Track unnecessary interventions, progress and paired-profile agreement. Keep positive/negative cases balanced in observable nuisance factors.

Paired contrast. Move an actor into the conflict corridor as a deliberate semantic change; do not mix it with the nuisance label.

Required profile contrast. Benign nominal policy passes; nuisance-pair profiles agree within declared tolerance.

Research diagnostic. Detects always-stop policies and shortcut learning. These are benchmark controls, not additional physical case types.

RC20 Incomplete library must remain unresolved

Site / intended evidence: Reuse RC04 or another witnessed positive. UNRESOLVED to R_REF when a held-out witness is admitted.

Mechanism. A restricted library contains only failing candidates, while a known successful causal continuation exists outside it.

CARLA construction. Use exactly the same physical scene and observation history. Hide the successful policy from the initial search or withhold a required world evaluation. Keep the complete witness in the evaluation-only record.

Parameters. Vary policy-library coverage or query budget, not physics. Give policy descriptors meaningful parameters so withheld policies are not merely arbitrary IDs.

Witness or refutation. Initially report U_LIBRARY only if that finite library is exhaustively refuted; otherwise UNRESOLVED. Once the successful policy is included and checked, record R_REF for the expanded class.

Accept only when. Never encode missing margins as zero or failure. Store library version and mask unresolved entries. Physical truth did not change when search coverage changed.

Paired contrast. Compare restricted and expanded libraries using one scenario-family split key.

Required profile contrast. No observed witness in the restricted record; a verified witness exists in the expanded record.

Research diagnostic. A direct test that labels capture evidence rather than planner weakness; also enables policy-transfer evaluation.

7. Reconstruct cases efficiently in CARLA

Use three reusable site types and calibrated motion primitives.

7.1 Pin the runtime and find sites once

Record the installed CARLA server build, matching Python API, map identifier/hash, vehicle blueprints and physics profile. Prior project notes mention multiple CARLA versions; this report does not assume which is running. Avoid a simulator upgrade solely to build this benchmark. No exact Town coordinates or actor assets have been verified here.

Create a small site registry: a straight constrained corridor, a two-lane straight segment with a recovery corridor, and a junction/occlusion or two-exit site. CARLA exposes map waypoints and road topology; OpenDRIVE can describe road networks when an existing suitable site is unavailable [9,10]. Prefer an already qualified site before creating a custom map.

7.2 Place actors in a local road frame

Define longitudinal s along travel and lateral d positive to the travel-right. Use the local tangent and right normal to construct transforms, then verify lane direction. CARLA uses x-forward, y-right, z-up, meters and degree-valued rotations [7]. A ROS convention may differ; explicitly test left/right and yaw conversions rather than assuming signs agree.

$$p(s,d) = p_{anchor} + s\,e_{forward} + d\,e_{right}.$$

The formula is a straight-site construction only. Curved roads require waypoint/Frenet geometry. Separate road clearance from actor-center spacing. Derive bumper gaps from world-space oriented footprints, including the vehicle bounding-box offset. CARLA exposes vehicle bounding boxes and their transformed vertices [8]. Record the site's permitted emergency corridor explicitly, including whether a shoulder is allowed.

7.3 Build reusable calibrated policies

Start with nominal lane following, straight braking plus hold, left/right bypass plus stabilization, a staged brake–change policy, an abort policy and one observation-triggered branch policy. These are finite reference policies, not a claim to cover every possible recovery. Store parameters, branch predicates, saturation behavior and terminal mode with each version.

Measure straight stopping, response delay, steering-rate response, lateral tracking and combined brake–steer behavior for the actual blueprint. CARLA supports runtime vehicle/wheel physics controls [11]; any altered profile needs recalibration. Preserve full physics for actors participating in decisive interactions. Do not establish feasibility by teleporting a vehicle along a candidate trajectory.

7.4 Use a single scenario harness

ScenarioRunner supports Python scenario definitions and trigger/behavior/criterion structure [12]. Reuse an existing compatible harness if available. The proposed JSON/YAML files describe our own case schema, not a built-in CARLA format. A compiler must resolve map sites, blueprints and policy implementations before launch. Do not add Autoware/ROS2 integration to the first mechanism test unless it is already stable and needed for the measured contract.

8. Timing, observation and actor contracts

A reproducible scene must also be a reproducible information history.

8.1 Simulation time is the experiment clock

CARLA's documented reproducibility setup uses synchronous stepping, a fixed simulation step, appropriate substepping and controlled world reloads. Only one client should advance the world. GPU sensor outputs can arrive later than their capture frames [4]. Pin these settings and run repeatability checks on the installed build; a seed alone is not proof of identical physical outcomes.

```
# Proposed pilot settings; verify on the installed CARLA build.
settings.synchronous_mode = True
settings.fixed_delta_seconds = 0.05
settings.substepping = True
settings.max_substep_delta_time = 0.01
settings.max_substeps = 5
# Required: fixed_delta_seconds <= max_substep_delta_time * max_substeps
```

Quantize configured delays to the tick grid and log the realized value. Observation capture, delivery, inference completion, command application and measured motion response need separate timestamps. Wall-clock sleeping is not a reliable simulated actuator-delay model. During a queued delay, apply the actual previous control rather than pausing ego motion.

8.2 Common histories and explicit information gates

For the first RC10/11 experiment, use a documented observation mask/delivery queue over simulator-derived structured observations. This deliberately tests decision making under partial observation, not visual detection accuracy. Later add RGB or other sensors with a physical occluder. CARLA sensor data contain frame and timestamp fields [6]; match them to the corresponding world snapshot before computing features or labels.

Verify identical allowed inputs and controller state before the reveal event in indistinguishable worlds. After reveal, only delivered observations may drive branching. A hidden actor's semantic tag, role name, blueprint, future coordinate or world index must not leak through a convenience feature. Recompute the compatible world set when the observation history changes.

8.3 Actor behavior must remain reactive and reproducible

Use project-owned finite-state actor controllers for decisive actors: bounded cruise, delayed braking, gap-triggered yielding and explicitly non-yielding modes. Key transitions to simulation time or observed geometric events. Define exogenous modes before the episode. Do not draw a new hidden intent independently at every tick unless that is the declared mechanism.

Traffic Manager can provide background traffic and supports deterministic seeding in synchronous mode; reseeding is needed after reload [5]. Keep uncontrolled background actors out of the initial labeling problem. Use matched actor mechanisms across ego alternatives, not recorder playback of unchanged actors. Physics-disabled hybrid traffic must not enter a case whose feasibility depends on its physical response.

9. Label-generation and acceptance protocol

Build expected contrasts first; promote labels only after independent checks.

9.1 Witness-first construction

For each positive target, first construct a complete successful policy in the reference model. Place or time the hazard relative to that executable response. For the negative counterpart, change one declared variable—distance, escape availability, reveal time or delay. The resulting intention remains provisional until evaluated. This is controlled scenario synthesis, not dataset mining.

9.2 Evidence-backed labeling

```
for each scenario family and allowed history:
    validate site, timing, observation and initial-state contracts
    for each policy in the declared library:
        for each compatible reactive world:
            execute a fresh matched continuation
            score hard margins and terminal behavior
            store witness / counterexample / unresolved evidence
    if a policy passes every declared world:
        record R_REF at the documented evidence scope
    elif every library policy has a valid counterexample:
        record U_LIBRARY (not U_MODEL)
    else:
        record UNRESOLVED
    attach a stronger U_MODEL or U_INFORMATION certificate only if supplied
```

For empirical CARLA runs, a repeat is a new matched episode with equivalent prehistory, not a presumed save-state restoration. Do not assume arbitrary hidden engine state can be cloned from actor poses alone. Store enough initialization, warm-up and controller-state information to re-establish the declared starting tolerance.

9.3 Validate geometry, timing and terminal behavior independently

Use timestamp-aligned oriented body geometry, a conservative swept check or a temporally refined collision calculation, and a separate permitted-road test. Discrete clear poses can hide between-tick intersection. Corroborate physical contact with the collision sensor; its events contain the other actor and impulse [6]. Lane invasion alone is not a universal road-safety verdict because permitted lane changes may cross markings.

Evaluate until the prescribed post-maneuver interval finishes. Keep model scores separate from evaluator scores. A mismatched tracker, late packet, missing collision stream, incorrect actor mode or unqualified vehicle profile makes the run invalid or unresolved, not evidence of unrecoverability.

9.4 Boundary and repeatability gates

Retain a safe interior anchor, a boundary band and a negative interior anchor for each applicable family. Use repeated executions at fixed settings, then perturb calibrated nuisance bounds. If signs change near the threshold, mark the band unresolved or report probabilistic performance separately. Never move the threshold after seeing held-out student results.

10. What the benchmark measures for our research

The case catalogue is a diagnostic instrument, not the method contribution itself.

10.1 Measure recovery profiles, not only collisions

The proposal trains policy-conditioned profiles and tests their usefulness after policies change [1, pp. 3–5]. Store each resolved margin, its failure component, supporting world, reference scope, and terminal mode. Keep unknown entries masked. Two scenes may both be recoverable but support different policies; RC03/04/07/15 are intended to expose that distinction.

Test	Evidence that matters
False admission	Among admitted resolved policy/history pairs, fraction refuted by the reference; also report the fraction of reference-negative pairs admitted.
False rejection	Among supported reference pairs, fraction rejected; report retained feasible actions and progress.
Profile fidelity	Margin error near the threshold, incorrect option-set changes and multi-step profile consistency.
Physical continuation	Contact, road departure, stabilization failure, secondary conflict, unnecessary intervention and task progress.
Information correctness	Pre-reveal action consistency, premature branching, history-sensitive decisions and late-packet handling.
Evidence integrity	Resolved fraction, invalid-run rate, search cost, teacher calls, executed policy/world continuations and repeatability.

10.2 Matched contrasts and model baselines

Compare prediction-only learning, an equal-architecture safety readout, a direct policy-conditioned recovery predictor, recovery supervision without the representation-distance term, and the proposed full method. For the closest method ablations, use the same labels and candidate policies. Measure the raw student separately from the unchanged external safety authority. Better labels or a better low-level controller must not be credited to the representation objective.

10.3 Prevent leakage and test transfer

Split complete counterfactual families before labeling/training: all distance neighbors, observation variants, renderings and policy-withholding variants of the same base instance stay together. Site reuse is allowed for a mechanism study but must be reported; reserve sites and actor laws for transfer. Policy-composition holdout requires withholding its outcome supervision, not only renaming a known controller.

Use paired independent episodes and training seeds. Report intervals at the independent episode/family level rather than treating every frame as an independent safety trial. Engineered critical-suite failure rates are conditional benchmark measurements, not estimates of ordinary-road crash frequency. The benchmark is successful as a tool if it finds that a simpler baseline already solves a proposed mechanism; do not weaken that baseline to manufacture an advantage.

11. Build priorities, compute and team interfaces

Construct a small audit suite before generating training-scale data.

11.1 Recommended construction order

Stage	Cases / deliverable	Completion criterion
Foundation	RC01, RC02, RC16, RC17	Stopping, geometry, timing and label scope agree in independently inspected traces.
Core mechanism	RC03, RC07, RC12, RC19, RC20	Different recovery profiles, post-maneuver effects and unresolved labels are visible without a learned model.
Compositional recovery	RC04, RC05, RC06, RC08, RC15, RC18	At least one audited positive witness per family; meaningful equal-data model comparisons are possible.
Partial observation	RC10, RC11, RC14	Information-tree reference and leakage tests pass; early/late evidence contrast is qualified.
Difficult negatives	RC09, RC13	Negative scope and predictor-approximation claims have defensible evidence, not only failed search.

A useful first batch is six selected contrasts × three parameter settings × three matched seeds = 54 physical instances. With eight policies and two compatible worlds each, a complete single pass requires 864 policy/world executions. These are arithmetic planning examples, not required sample sizes or a statistical power calculation. Count repetitions, setup and post-maneuver time separately.

11.2 Save effort without corrupting labels

Cache calibrated motion profiles and site geometry; reuse the same family initializer; batch low-dimensional reference evaluation; rerender selected instances only after labels are audited. Search near a known mechanism boundary rather than perturbing all variables at once. Keep future training examples separate from test-family construction. The first milestone is a trustworthy label generator, not a large neural network.

11.3 Suggested work division

Scenario/evidence lead: owns the observation contract, reference actors, policy qualification, swept geometry, terminal test and evidence audit. **Model/evaluation lead:** owns profile schema, loading/masking, equal-data baselines, split integrity, uncertainty analysis and cost accounting. Jointly inspect at least one positive, one refuted policy and one unresolved instance before parallel expansion.

Existing CARLA/ROS2 and recovery-checking code may be reusable, as stated in the proposal [1, p. 6], but this report has not audited those repositories. Do not require a new full-stack integration for the initial finite reference tests, and do not change an existing production/authority boundary as a side effect of benchmark generation.

Decision rule for expansion

Advance only when labels are supported and reproducible. A template that repeatedly violates its intended mechanism must be revised or excluded, not relabeled to fit the story. Publish the exact policy/world contract with every result so that benchmark coverage and remaining gaps are visible.

12. Reproducible case manifest and stored evidence

The accompanying YAML is a project schema; an adapter is still required.

Keep the scenario specification, actual execution record and learning example separate. Intentions belong in the specification; measured margins and evidence belong in execution records; hidden metadata must not enter model features.

```yaml
case_id: RC01
family_id: corridor_stop_base_001
schema_version: "0.1"
status: NOT_EXECUTED
site:
  map: null                 # resolve on the installed simulator
  site_id: straight_corridor_01
contract:
  observation_mode: masked_structured_history
  world_set_id: stationary_blocker_no_rear_v1
  policy_library_id: calibrated_recovery_v1
  physics_profile_id: null  # must be measured and versioned
  fixed_delta_seconds: 0.05
  safety_threshold: 0.0
construction:
  speed_mps: 12.0
  gap_rule: measured_stop_distance_plus_reserve
  reserve_m: 2.0
  lateral_escape: false
terminal:
  mode: protected_hold
  post_maneuver_seconds: 4.0
intended_outcome: R_REF
actual_outcome: null
split_group: corridor_stop_base_001
required_checks:
  - world_history_compatibility
  - applied_control_and_latency
  - swept_footprint_and_road
  - complete_terminal_continuation
  - all_declared_worlds_evaluated
```

Additional mandatory execution fields: resolved transforms and blueprint IDs; actual geometry/physics values; every policy descriptor; capture/delivery/control/response timestamps; per-world raw and normalized margins; collision and road events; witness or refutation traces; policy/world coverage; termination cause; seed; code version; invalidity reasons; checksums and split group.

For counterfactual siblings, preserve a parent family ID and declare the single changed variable. For observation-only variants, log both the unchanged physical world and the changed information channel. For RC20, change the library/evidence field without pretending to change the physical scenario.

References and final specification boundary

Source-derived foundations are distinguished from the proposed catalogue.

[1] Yige Wei. Recoverability-Preserving World Models for Safety-Critical Imitation Learning. Extended draft, 28 September 2026, 6 pages. User-provided file: 01_recoverability_preserving_world_models(1).pdf. Core basis: Sections 4–6, pp. 2–5; existing/new-work boundary: Section 10, p. 6.

[2] T. Fraichard and H. Asama. Inevitable collision states — a step towards safer robots? Advanced Robotics 18(10), 1001–1024, 2004. DOI: 10.1163/1568553042674662.
https://inria.hal.science/inria-00182063

[3] Z. Zhang and J. F. Fisac. Safe Occlusion-aware Autonomous Driving via Game-Theoretic Active Perception. Robotics: Science and Systems, 2021.
https://arxiv.org/abs/2105.08169

[4] CARLA documentation. Synchrony and time-step. Fixed stepping, synchronization, substeps and sensor-delay notes.
https://carla.readthedocs.io/en/latest/adv_synchrony_timestep/

[5] CARLA documentation. Traffic Manager. Deterministic mode, seeding and hybrid-physics behavior.
https://carla.readthedocs.io/en/latest/adv_traffic_manager/

[6] CARLA documentation. Sensors reference. Collision events and timestamped sensor output.
https://carla.readthedocs.io/en/latest/ref_sensors/

[7] CARLA documentation. Coordinates and transformations. Coordinate axes, distances and rotation units.
https://carla.readthedocs.io/en/latest/coordinates/

[8] CARLA documentation. Python API reference. Actor bounding boxes, controls and world-space vertices.
https://carla.readthedocs.io/en/latest/python_api/

[9] CARLA documentation. Maps. Waypoints, road/lane identifiers and topology.
https://carla.readthedocs.io/en/latest/core_map/

[10] CARLA documentation. ASAM OpenDRIVE. Standalone road-network support.
https://carla.readthedocs.io/en/latest/adv_opendrive/

[11] CARLA documentation. Control and monitor vehicle physics. Runtime vehicle and wheel parameters.
https://carla.readthedocs.io/en/latest/tuto_G_control_vehicle_physics/

[12] CARLA ScenarioRunner documentation. Create a new scenario. Scenario behavior and evaluation structure.
https://scenario-runner.readthedocs.io/en/latest/creating_new_scenario/

Public documentation checked 15 October 2026. Documentation describes interfaces, not qualification of the local runtime. Pin matching versioned documentation after the installed build is resolved. The mathematical refinements, numerical illustrations, case cards, prioritization and manifest in this report are proposed here. No case is presented as an executed CARLA result, a certified vehicle capability or a frequency estimate for natural driving.